



MesoProbe

THE NEXT GENERATION OF
**MESO-SCALE
MECHANICAL TESTING**



NANOMETER
PRECISION



INTEGRATED
DIC



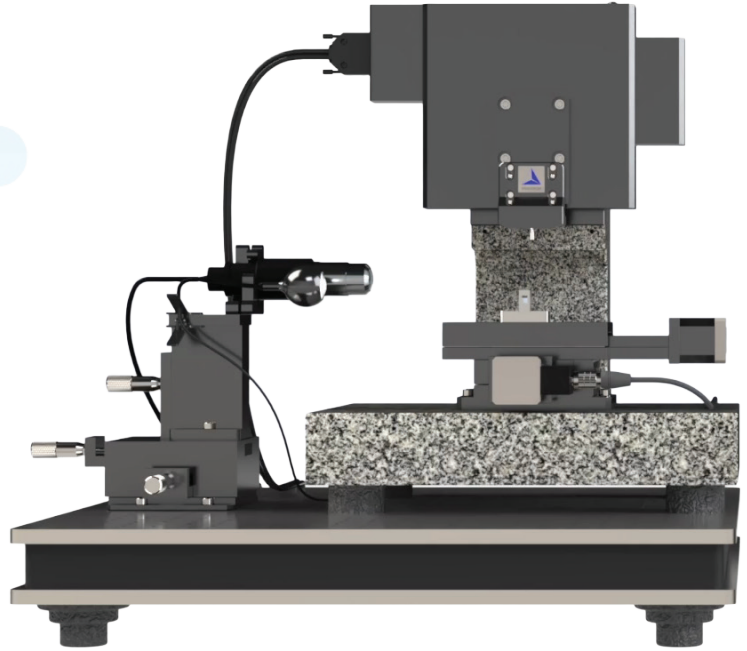
HIGH
THROUGHPUT

INDUSTRON TECHNICAL SERVICES PVT. LTD.

www.industronnano.com

WHY CHOOSE MesoProbe

MesoProbe is designed to bridge the gap between nano and macro testing—enabling accurate mechanical characterization on small samples with bulk-relevant insights.



THE GAP IN MECHANICAL TESTING



NANO SCALE

High resolution
Very small test volumes
Limited representation of
bulk behavior

1 nm – 10 μ m



MESO SCALE

The Sweet Spot

Bridges micro and macro
Small sample volume
High-throughput data
Integrated DIC
High-temperature testing
Multiple test modes

10 μ m – 5 mm



MACRO SCALE

Bulk properties
Large sample volume
Low spatial resolution

> 5 mm

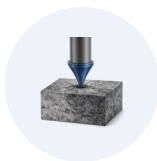


HOW IT WORKS



01

Small Sample
Minimal material
requirement



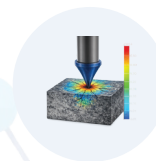
02

Mechanical Loading
Indentation, Compression,
Tensile, or Bending



03

High Resolution Imaging
Real-time in-situ optical
observation



04

Digital Image Correlation
Full-field strain mapping
with high accuracy



05

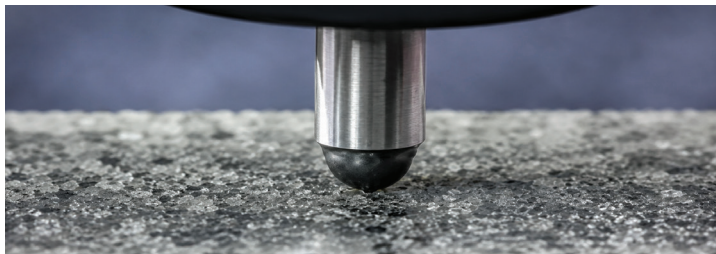
Mechanical Properties
Stress, Strain, Modulus,
Creep, Fatigue & more

ONE PLATFORM. MULTIPLE EXPERIMENTS.

MesoProbe supports a wide range of mechanical tests with integrated Digital Image Correlation for full-field strain analysis and high-throughput data generation.

EXPERIMENTS

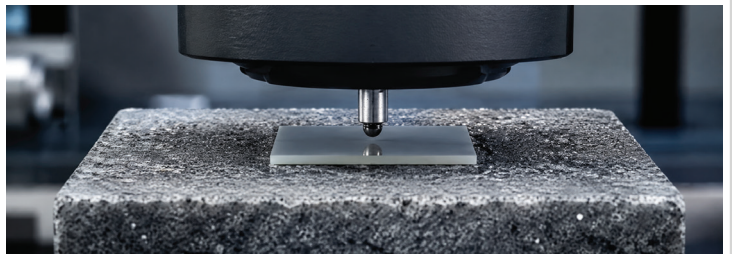
1. INDENTATION



Indentation Test using a Spherical Tip

- Hardness
- Elastic Modulus
- Load-Displacement Analysis
- Depth Sensing Indentation

2. COMPRESSION



Compression Test on a Soft Material Matrix

- Polymers, Hydrogels, Rubber, Foams
- Soft Materials Testing
- Low Force Sensitivity (μN Level)

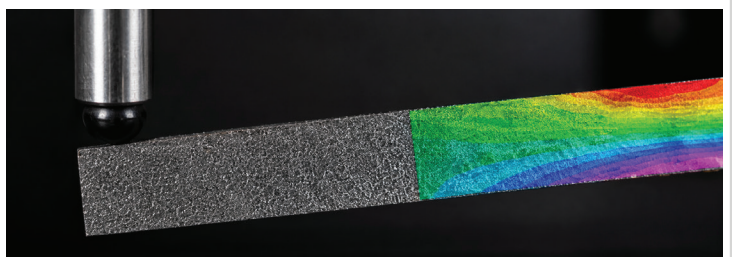
3. THREE-POINT BENDING



Three Point Bending Test on Soft Material Matrix with Strain Data from DIC Analysis data Overlaid

- Young's Modulus
- Stress & Strain Mapping
- Full-field Strain from DIC
- Fracture & Failure Analysis

4. CANTILEVER BENDING



Cantilever Bending Test on Aluminium Sample with Strain Data from DIC Analysis data Overlaid

- Creep & Fatigue Analysis
- Strain Evolution
- High Throughput Creep Testing
- Ideal for Limited Sample Material

INDUSTRIES & RESEARCH AREAS



AUTOMOTIVE



AEROSPACE



BATTERY
MATERIALS



THIN FILMS &
COATINGS



SEMICONDUCTORS
& MEMS



BIOMATERIALS &
MEDICAL DEVICES



EDUCATION &
RESEARCH LABS

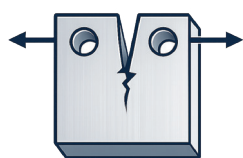
TECHNICAL SPECIFICATIONS

SPECIFICATION	DETAILS
 Maximum Actuation Load	20 N
 Maximum Displacement	60 mm
 Displacement Resolution	1 nm
 Camera Resolution	4024 px × 3036 px
 Temperature Capability	Up to 600°C
 Motorized Translation Stages	X-axis: 150 mm Y-axis: 50 mm
 Optics	0.2X (1X / 5X / 10X Optional)

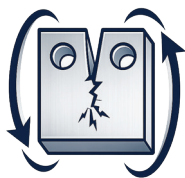


SMALL SAMPLES.
BIG INSIGHTS.

TEST MODES



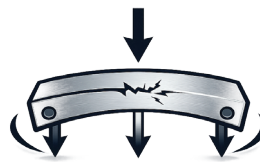
FRACTURE



FATIGUE



CREEP



BENDING



TENSILE

CONTACT US

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Scan to Explore
MesoProbe
 in Action